

Atividade de Álgebra Booleana - 2

1 - a) $A + AB'$

$$A + AB = A$$

$$A$$

b) $A'BC + BC$

$$BC(A' + 1)$$

$$BC(1)$$

$$BC$$

c) $(A + B)(A + B')$

$$A + BB'$$

$$A + 0$$

$$A + 0 = A$$

$$A$$

d) $AB + A'B + AB'$

$$AB + A'B$$

$$B(A + A')$$

$$B(1) = B$$

$$B + AB'$$

$$B = B(A + A')$$

$$B + AB + AB$$

$$A(B + B')$$

$$A$$

e) $(A + B + C)(A + B')(A' + C)$

$$(A + (B + C)(A' + C))$$

$$B'B' + B'C$$

$$0 + B'C$$

$$(A + B'C)(A' + C)$$

$$AA' + AC + A'B'C + B'CC$$

$$0 + AC + A'B'C + B'C$$

$$B'C(A' + 1)$$

$$AC + B'C$$

$$C(A + B')$$

2 - $AB + BC(B + C)$

Simplifique a expressão:

$$AB + BCB + BCC$$

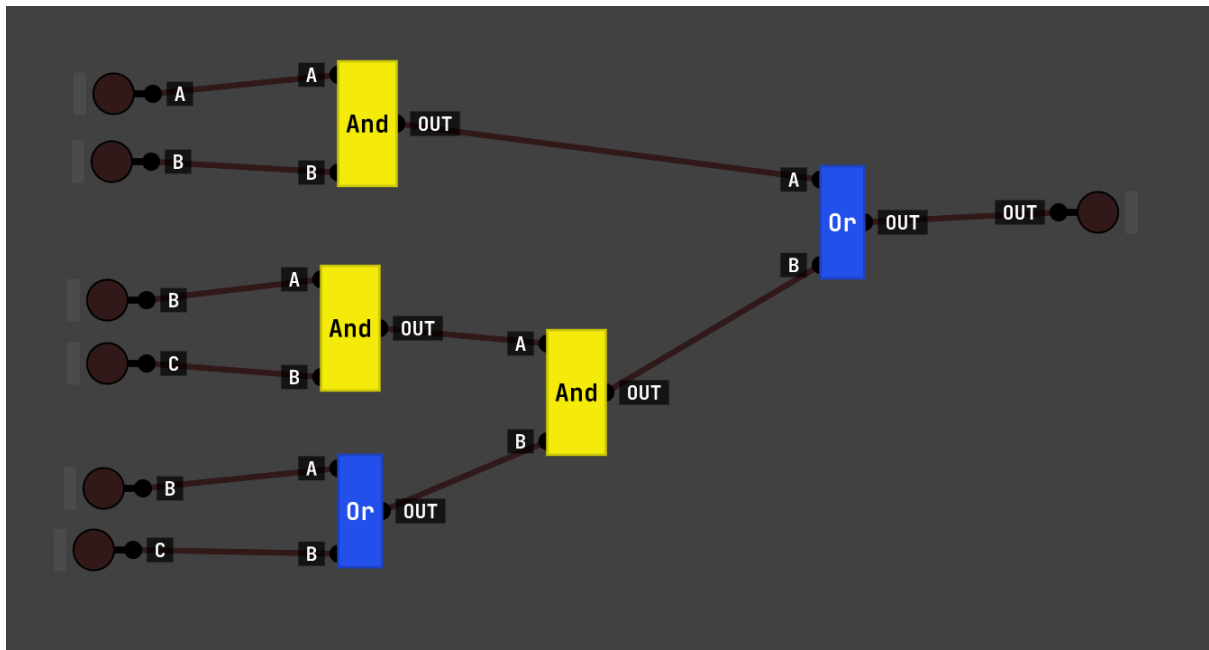
$$AB + BC + BC$$

$$AB + BC$$

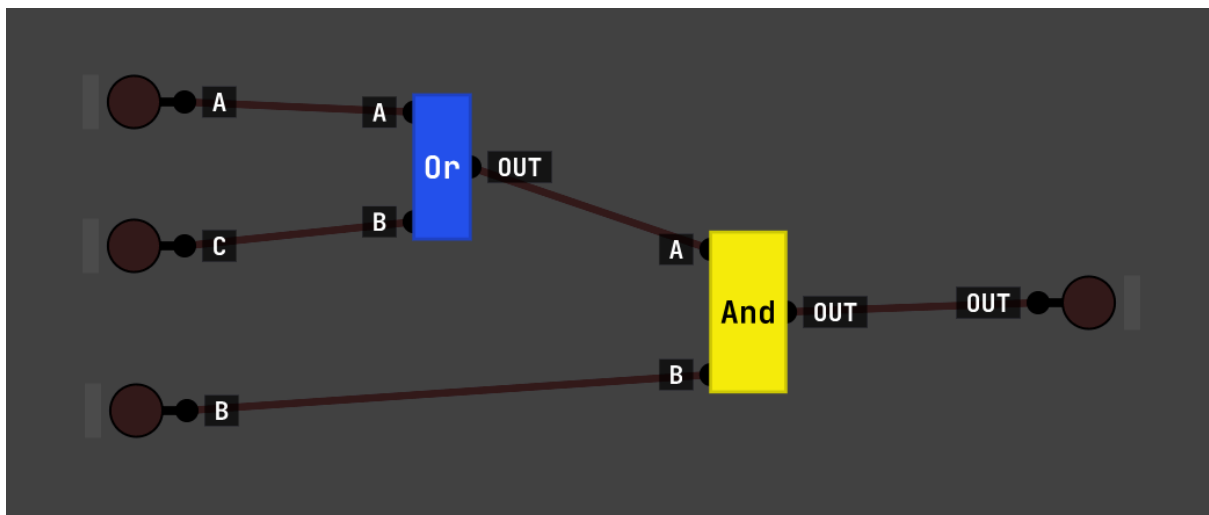
$$B(A + C)$$

Desenhe o circuito original e o simplificado:

ORIGINAL:



SIMPLIFICADO:



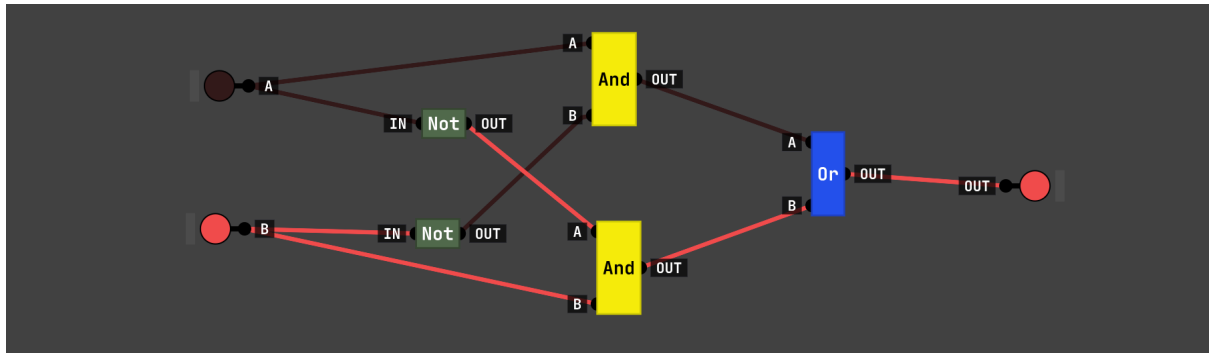
Compare o número de portas utilizadas:

Circuito original: 5 portas

Circuito simplificado: 2 portas

3 - $A \oplus B$

$$A'B + AB'$$



4 - $A \oplus B = A'B + AB'$

A	B	XOR
0	0	0
0	1	1
1	0	1
1	1	0

5 - a) $(A + B'C)'$

$$(A'(B'C'))'$$

$$(B'C')' = B + C$$

$$A'(B + C)$$

b) $(A' + B')' (A + C)'$

$$(A' + B')$$

$$(A')'B'$$

$$(A')' = A$$

$$AB'$$

$$(A + C)'$$

$$A'C'$$

$$AB' \cdot A'C'$$

$$AA'B'C'$$

$$AA = 0$$

$$c) (A'B' + C)'$$

$$(A'B')'C'$$

$$(A'B')'$$

$$(A')' + (B')'$$

$$A + B$$

$$(A + B)C'$$

7 - SOD

$$A'B'C + AB'C' + ABC$$

$$A'(B'C + BC') + A(B'C' + BC)$$

$$A'(B \oplus C) + A(B \oplus C)'$$

$$A'X + AX'$$

$$A \oplus X$$

$$A \oplus B \oplus C$$

POS

$$(A'B'C)'$$

$$A + B + C$$

$$(A + B + C)(A + B' + C')(A' + B + C')(A' + B' + C)$$

$$AA + B' + AC'$$

$$A + AC'$$

$$A + AB + BB' + BC'$$

$$A + BC' + B'C$$

$$A + B \oplus C$$

$$(A' + B + C')(A' + B' + C)$$

$$A' + A'B + A'C$$

$$A' + A'B + B'B + BC$$

$$A + (B \oplus C) \cdot A' + BC + BC'$$

$$A + B \oplus C \cdot A' + (B \oplus C)'$$

$$(A + X)(A' + X')$$

$$AA' + AX' + A'X + XX'$$

$$0 + AX' + AX' + 0$$

$$AX' + A'X$$

$$A \oplus X$$

$$A \oplus B \oplus C$$

8 -

A	B	C	pelo menos 2
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

SOP:

$A'BC + AB'C + ABC' + ABC$

$BC(A' + 1) + AB'C + ABC'$

$B(C + AC') + AB'C$

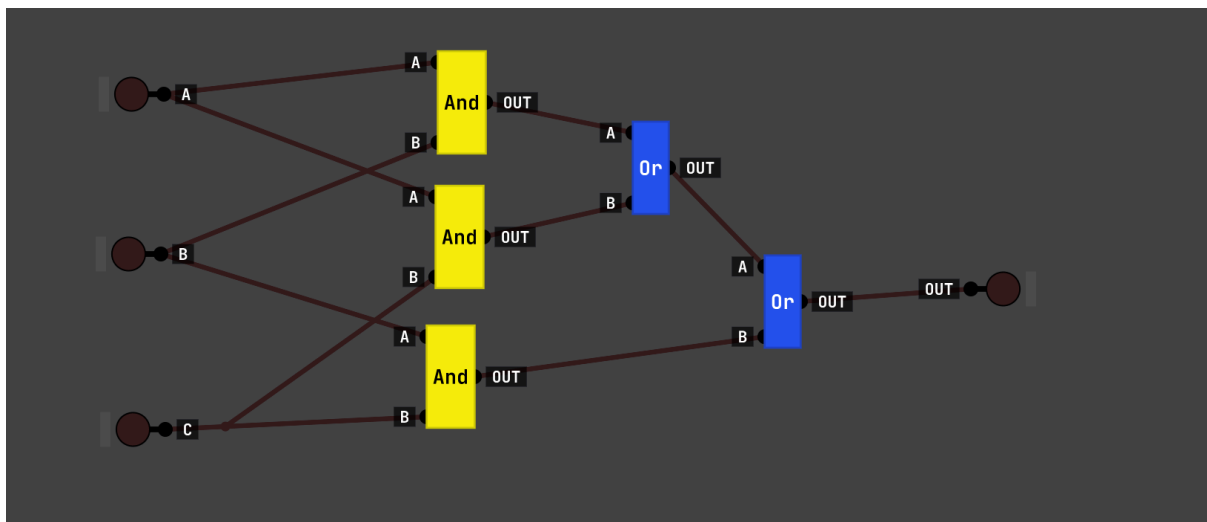
$AB + BC + AB'C$

$A(B + B'C) + BC$

$A(B + C) + BC$

$AB + AC + BC$

$AB + AC + BC$ (A e B) ou (A e C) ou (B e C)



9 -

$$(B + C' + D)(A' + B) = \text{POS}$$

$$AB'C' + A'BC = \text{SOP}$$

$$(X + Y' + Z')(Y' + Z)(X' + Y) = \text{POS}$$

$$M'N'O' + MNO' + MN'O = \text{SOP}$$

$$(X + (Y + Z)')(Y + Z')' = \text{POS}$$

$$(ABC)' + AB'C = \text{SOP}$$